Nagar Yuwak Shikshan Sanstha's

# Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

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## Department of Computer Technology

**Vision of the Department**

*To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.*

**Mission of the Department**

*To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.*

**Session 2026-2027(ODD)**

| **Vision:** Dream of where you want. | **Mission:** Means to achieve Vision |
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**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

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| PEO1 | **Preparation** | **P: Preparation** | **Pep-CL abbreviation pronounce as Pep-si-lL easy to recall** |
| PEO2 | **Core Competence** | **E: Environment (Learning Environment)** | |
| PEO3 | **Breadth** | **P: Professionalism** | |
| PEO4 | **Professionalism** | **C: Core Competence** | |
| PEO5 | **Learning Environment** | **L: Breadth (Learning in diverse areas)** | |

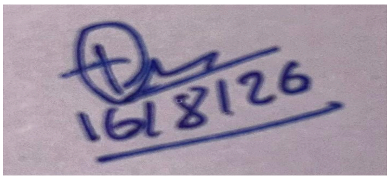
**Program Outcomes (PO):** (statements that describe what a student should be able to do and know by the end of a program)

**Keywords of POs:**
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

**PSO Keywords:** Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

**Integrity:** I will adhere to the Laboratory Code of Conduct and ethics in its entirety.




**Name and Signature of Student and Date**
(Signature and Date in Handwritten)

| Session | 2026-27 (ODD) | Course Name | LAB: Software Engineering |
|---|---|---|---|
| Semester | 7th | Course Code | 23CT1704 |
| Roll No | 83 | Name of Student | Trupal Tagade |

| Practical Number | 3 |
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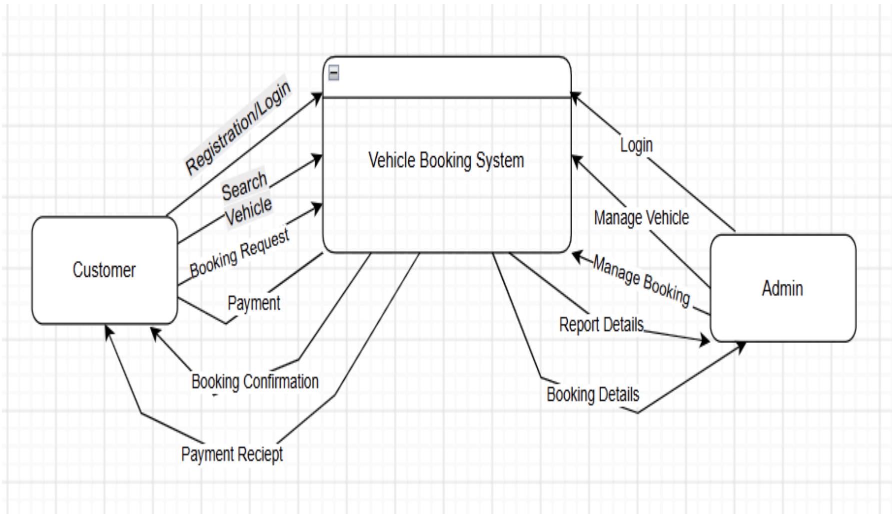
| Aim | **To study and draw the Data Flow Diagrams (DFDs) at Level 0 (Context Diagram) and Level 1 for the Vehicle Booking System using StarUML.** |
| Theory<br>(100 words) | A **Data Flow Diagram (DFD)** is a graphical representation that shows how **data moves through a system**. It focuses on data inputs, outputs, processes, data stores, and data transformations.<br><br>**Key Components of DFD**<br><br>1. **External Entity:** A person or external system that interacts with the system, such as **Customer** or **Admin**.<br>2. **Process:** An operation that transforms input data into output data, such as **Check Vehicle Availability**.<br>3. **Data Store:** A place where data is stored, such as **Vehicle Database** or **Booking Database**.<br>4. **Data Flow:** Arrows showing the movement of data between entities, processes, and data stores. |
| Procedure and Execution<br><br>(100 Words) | **<u>Level 0 DFD (Context Diagram)</u>** |

A Level 0 DFD, also called the Context Diagram, provides a high-level view of the entire system.

**Characteristics:**

- Represents the whole system as one single process.
- Shows only external entities interacting with the system.
- Does not show internal processes or data stores.
- Used to understand the overall scope of the system.




## Level 0 DFD

**Level 1 DFD (System Breakdown)**

A Level 1 DFD zooms into the single process from Level 0, exploding it into its major functional sub-processes.

**Characteristics:**

1. **Multiple Sub-Processes:** Decomposes Process 0 into distinct steps (e.g., 1.0 Process Order, 2.0 Manage Inventory, 3.0 Generate Invoice).

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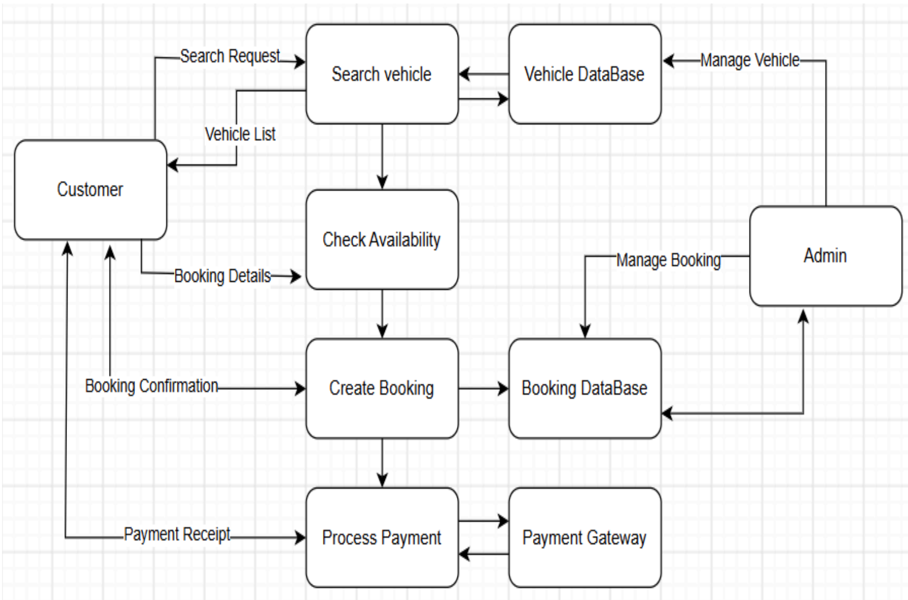
| | 2. **Reveals Data Stores:** Shows internal databases, tables, or file systems for the first time.<br>3. **Data Flow Preservation:** Every data flow coming from an external entity in Level 0 must still appear in Level 1 (a rule known as *balancing*).<br><br><br>**Level 1 DFD** |
| Link of student Github profile where lab assignment has been uploaded | [23070254-sys](23070254-sys) |
| Conclusion | The Vehicle Booking System DFD successfully represents how data flows through the system. The Level 0 DFD gives |

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| | an overall view of the system, while the Level 1 DFD shows the main processes such as vehicle search, availability checking, booking, and payment. It helps in understanding the system clearly and efficiently. |
| Date | **10/8/2026** |